```
name : asmaa ahmed maryah
id: 811 634 546
dataset: https://www.kaggle.com/datasets/amar5693/screen-time-sleep-and-stress-analysis-dataset
what data set talk about:
This dataset contains 50,000 user records analyzing smartphone usage behavior and its relationship with productivity, sleep patterns, and stress levels

# print top 5 rows without first row(header)
import pandas as pd
df = pd.read_csv( 'Smartphone_Usage_Productivity_Dataset_50000.csv' ,
skiprows=1 )
df.head()
```

```
   U1  58    Male    Professional  Android   1.3  6.7  6  8.8  4  42
1   8.7
0  U2  25    Male    Professional  Android   1.2  1.5  5  6.4  1  51
3   5.1
1  U3  19    Male         Student      iOS   5.3  5.7  5  9.0  4  14
5   6.3
2  U4  35  Female  Business Owner      iOS   5.8  2.5  2  5.7  3  36
6  12.8
3  U5  33    Male      Freelancer  Android   7.9  1.3  4  5.7  3  37
5   9.9
4  U6  32  Female         Student  Android  10.9  4.2  9  6.3  7  34
5   3.6
```

```
# print the last 10 rows
df.tail(10)
```

```
      User_ID  Age  Gender      Occupation Device_Type
Daily_Phone_Hours  \
49990  U49991   24   Other      Freelancer         iOS
6.1
49991  U49992   42  Female  Business Owner     Android
11.1
49992  U49993   34    Male    Professional         iOS
10.1
49993  U49994   38   Other      Freelancer     Android
2.2
49994  U49995   34    Male         Student     Android
4.6
49995  U49996   44    Male  Business Owner     Android
5.9
49996  U49997   42   Other  Business Owner     Android
2.9
49997  U49998   27  Female      Freelancer         iOS
1.4
49998  U49999   41  Female  Business Owner         iOS
```

```
8.9
49999  U50000   46   Other      Freelancer     Android
8.8

       Social_Media_Hours  Work_Productivity_Score  Sleep_Hours
Stress_Level  \
49990                 7.4                        9          6.8
6
49991                 1.4                        6          8.2
7
49992                 4.1                        3          7.3
8
49993                 3.1                        1          4.1
10
49994                 7.0                        4          4.4
3
49995                 5.4                        5          6.6
1
49996                 7.4                        9          6.3
2
49997                 2.5                        4          6.7
9
49998                 3.0                        6          5.5
2
49999                 2.8                        3          5.4
1

       App_Usage_Count  Caffeine_Intake_Cups
Weekend_Screen_Time_Hours
49990               23                     1
2.6
49991               33                     5
8.6
49992               36                     3
5.7
49993               18                     4
4.0
49994               44                     3
7.4
49995               11                     5
3.0
49996               20                     4
6.2
49997               39                     4
5.1
49998               51                     0
9.2
49999               54                     5
6.9
```

```python
# load dataset
import pandas as pd
df = pd.read_csv( 'Smartphone_Usage_Productivity_Dataset_50000.csv' )

# inof about dataset
df.info
```

```
<bound method DataFrame.info of        User_ID  Age  Gender
Occupation Device_Type  Daily_Phone_Hours  \
0           U1   58    Male    Professional     Android
1.3
1           U2   25    Male    Professional     Android
1.2
2           U3   19    Male         Student         iOS
5.3
3           U4   35  Female  Business Owner         iOS
5.8
4           U5   33    Male      Freelancer     Android
7.9
...        ...  ...     ...             ...         ...
...
49995  U49996   44    Male  Business Owner     Android
5.9
49996  U49997   42   Other  Business Owner     Android
2.9
49997  U49998   27  Female      Freelancer         iOS
1.4
49998  U49999   41  Female  Business Owner         iOS
8.9
49999  U50000   46   Other      Freelancer     Android
8.8

       Social_Media_Hours  Work_Productivity_Score  Sleep_Hours
Stress_Level  \
0                     6.7                        6          8.8
4
1                     1.5                        5          6.4
1
2                     5.7                        5          9.0
4
3                     2.5                        2          5.7
3
4                     1.3                        4          5.7
3
...                   ...                      ...          ...
...
49995                 5.4                        5          6.6
1
49996                 7.4                        9          6.3
2
```

```
49997                  2.5                         4          6.7
9
49998                  3.0                         6          5.5
2
49999                  2.8                         3          5.4
1

       App_Usage_Count  Caffeine_Intake_Cups
Weekend_Screen_Time_Hours
0                    42                     1
8.7
1                    51                     3
5.1
2                    14                     5
6.3
3                    36                     6
12.8
4                    37                     5
9.9
...                 ...                   ...                        ..
.
49995                11                     5
3.0
49996                20                     4
6.2
49997                39                     4
5.1
49998                51                     0
9.2
49999                54                     5
6.9

[50000 rows x 13 columns]>

# print column names
df.columns

Index(['User_ID', 'Age', 'Gender', 'Occupation', 'Device_Type',
       'Daily_Phone_Hours', 'Social_Media_Hours',
'Work_Productivity_Score',
       'Sleep_Hours', 'Stress_Level', 'App_Usage_Count',
       'Caffeine_Intake_Cups', 'Weekend_Screen_Time_Hours'],
      dtype='object')

# made a subset of columns and select 5 random rows
df_1 = df[['User_ID' , 'Age' , 'Gender']].sample(n=5)
df_2 = df[['User_ID' , 'Age' , 'Gender']].sample(n=5)
df_3 = df[['User_ID' , 'Age' , 'Gender']].sample(n=5)
```

```python
# print created subsets
print(f"{df_1}\n\n{df_2}\n\n{df_3}")
```

```
      User_ID  Age  Gender
45048  U45049   22  Female
32355  U32356   47   Other
36284  U36285   27  Female
12072  U12073   20  Female
31163  U31164   58    Male

      User_ID  Age  Gender
43910  U43911   19   Other
4223    U4224   41  Female
25429  U25430   30    Male
26533  U26534   45   Other
47276  U47277   38   Other

      User_ID  Age  Gender
27655  U27656   28   Other
32779  U32780   50  Female
12177  U12178   46    Male
35226  U35227   48   Other
10315  U10316   56  Female
```

```python
# horizontal concatination
horizontal_concatination = pd.concat( [df_1 , df_2 , df_3] ,axis=1 )
horizontal_concatination
```

```
      User_ID   Age  Gender User_ID   Age  Gender User_ID   Age
Gender
45048  U45049  22.0  Female     NaN   NaN     NaN     NaN   NaN
NaN
32355  U32356  47.0   Other     NaN   NaN     NaN     NaN   NaN
NaN
36284  U36285  27.0  Female     NaN   NaN     NaN     NaN   NaN
NaN
12072  U12073  20.0  Female     NaN   NaN     NaN     NaN   NaN
NaN
31163  U31164  58.0    Male     NaN   NaN     NaN     NaN   NaN
NaN
43910     NaN   NaN     NaN  U43911  19.0   Other     NaN   NaN
NaN
4223      NaN   NaN     NaN   U4224  41.0  Female     NaN   NaN
NaN
25429     NaN   NaN     NaN  U25430  30.0    Male     NaN   NaN
NaN
26533     NaN   NaN     NaN  U26534  45.0   Other     NaN   NaN
NaN
47276     NaN   NaN     NaN  U47277  38.0   Other     NaN   NaN
NaN
```

```
27655      NaN   NaN      NaN      NaN   NaN      NaN  U27656  28.0
Other
32779      NaN   NaN      NaN      NaN   NaN      NaN  U32780  50.0
Female
12177      NaN   NaN      NaN      NaN   NaN      NaN  U12178  46.0
Male
35226      NaN   NaN      NaN      NaN   NaN      NaN  U35227  48.0
Other
10315      NaN   NaN      NaN      NaN   NaN      NaN  U10316  56.0
Female
```

```
# vertical oncatination
vertical_oncatination = pd.concat( [df_1 , df_2 , df_3], axis=0 )
vertical_oncatination
```

```
      User_ID  Age  Gender
45048  U45049   22  Female
32355  U32356   47   Other
36284  U36285   27  Female
12072  U12073   20  Female
31163  U31164   58    Male
43910  U43911   19   Other
4223    U4224   41  Female
25429  U25430   30    Male
26533  U26534   45   Other
47276  U47277   38   Other
27655  U27656   28   Other
32779  U32780   50  Female
12177  U12178   46    Male
35226  U35227   48   Other
10315  U10316   56  Female
```

```
# select the first 5 eows
df_1 = df[['User_ID']][:5]
df_2 = df[['Age']][:5]
df_3 = df[['Gender']][:5]

df_1 , df_2 , df_3
```

```
(  User_ID
 0      U1
 1      U2
 2      U3
 3      U4
 4      U5,
    Age
 0   58
 1   25
 2   19
 3   35
```

```
 4   33,
    Gender
 0    Male
 1    Male
 2    Male
 3  Female
 4    Male)
```

```python
# horizontal concatination
horizontal_concatination = pd.concat([df_1 , df_2 , df_3] ,axis=1)
horizontal_concatination
```

```
  User_ID  Age  Gender
0      U1   58    Male
1      U2   25    Male
2      U3   19    Male
3      U4   35  Female
4      U5   33    Male
```

```python
# statical summary about dataset
df.describe
```

```
<bound method NDFrame.describe of       User_ID  Age  Gender
Occupation Device_Type  Daily_Phone_Hours  \
0          U1   58    Male    Professional     Android
1.3
1          U2   25    Male    Professional     Android
1.2
2          U3   19    Male         Student         iOS
5.3
3          U4   35  Female  Business Owner         iOS
5.8
4          U5   33    Male      Freelancer     Android
7.9
...       ...  ...     ...             ...         ...
...
49995  U49996   44    Male  Business Owner     Android
5.9
49996  U49997   42   Other  Business Owner     Android
2.9
49997  U49998   27  Female      Freelancer         iOS
1.4
49998  U49999   41  Female  Business Owner         iOS
8.9
49999  U50000   46   Other      Freelancer     Android
8.8

       Social_Media_Hours  Work_Productivity_Score  Sleep_Hours
Stress_Level  \
0                     6.7                        6          8.8
```

```
4
1                      1.5                           5           6.4
1
2                      5.7                           5           9.0
4
3                      2.5                           2           5.7
3
4                      1.3                           4           5.7
3
...                    ...                         ...           ...
...
49995                  5.4                           5           6.6
1
49996                  7.4                           9           6.3
2
49997                  2.5                           4           6.7
9
49998                  3.0                           6           5.5
2
49999                  2.8                           3           5.4
1

       App_Usage_Count  Caffeine_Intake_Cups
Weekend_Screen_Time_Hours
0                    42                     1
8.7
1                    51                     3
5.1
2                    14                     5
6.3
3                    36                     6
12.8
4                    37                     5
9.9
...                 ...                   ...                        ..
.
49995                11                     5
3.0
49996                20                     4
6.2
49997                39                     4
5.1
49998                51                     0
9.2
49999                54                     5
6.9

[50000 rows x 13 columns]>
```

```python
# catigorical attributes
Device_Type = pd.DataFrame({'Device_Type':
df['Device_Type'].value_counts()})
Device_Type
```

	Device_Type
Device_Type	
Android	25080
iOS	24920